

Advantage Contracts for Anonymous DHT Evaluations: Stealth Audits, Leakage Discovery at Scale, and Practical Padding Filters

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Abstract

Anonymous DHTs and overlay lookups are typically evaluated with active probes: canary lookups, synthetic workloads, and measurement traffic. A practical adversary can *detect* these audits and behave differently under evaluation than in deployment (“sandbagging”), biasing published anonymity and reliability claims. We propose an *advantage contract* that makes evaluation bias explicit and controllable: bound the audit-live distinguishability (TV advantage) of the observable traces used by the evaluator. Under this contract, any bounded evaluation metric (risk, leakage proxy, success rate) cannot shift by more than the advantage budget. We then make the contract operational with (i) anytime-valid monitoring of detectability via e-processes, (ii) scalable slice discovery with e-values and e-BH, and (iii) practical stealth mechanisms (commit-reveal probe selection with public randomness) plus padding filters that minimize overhead subject to an advantage constraint. We include a lightweight Kademlia-flavored trace case study showing how detectable audit infrastructure induces audit-live bias and how stealth reduces both.

Compilation note. This paper is brief-by-design. For extended deployed-evidence pointers and additional bibliography, see the related work map.[37]

Scope. We study *evaluation awareness*: the system under test may condition on whether traffic is an audit, inducing an audit-live distribution shift. We package this shift as a published *advantage contract* that upper-bounds audit detectability on the released trace schema.

Imports. We import the trace-schema / exposure-tier vocabulary and the TV/padding compiler toolchain from the companion series notes.[1, 2] For stopping-time leakage and deadline adaptivity, we import the stop-time addendum.[3] For certificate mechanics and retry discipline, we import the notarized certificates note.[17] Exposure-normal-form identity and interface/coarsening semantics are imported from the trace-interface note.[23]

Exports. We export (i) an advantage contract (TV distinguishability) linking audit detectability to bounded bias in any reported metric, (ii) anytime-valid monitoring of detectability via e-processes and scalable slice discovery (e-values, e-BH), and (iii) practical stealth mechanisms (commit-reveal probe selection with public randomness) and padding filters that reduce detectability subject to an advantage budget.

Artifact policy. This archive is source-only; supporting artifacts (datasets, plots, scripts, and tooling) are available on request.[36]

1 Series context and scope

This paper connects to two prior themes in the series: (1) *trace-based* anonymity objectives and TV leakage bounds, and (2) padding/scheduling compilers and transport-style certificates [1, 2]. We do not re-prove these foundations. For stopping-time leakage and deadline adaptivity on the privacy side, see the series addendum [3]. Our focus is *evaluation awareness*: the system under test may condition on whether traffic is an audit. This is analogous to overfitting a test/holdout set under adaptive reuse: if a system can infer which queries are audits, it can “fit” to the evaluator. The statistics literature studies such adaptivity and mechanisms for safe holdout reuse [24, 25].

Series vocabulary (shared). We use a small, consistent vocabulary across the series to keep papers non-redundant. A *plan id* names an evaluation definition (trace schema, attempt unit, within-attempt library/correction, and spending schedule); changing any evaluation knob implies a new plan id. An *epoch* (or window) is the declared validity interval of a plan (e.g., a wall-clock period or a beacon-derived id); changing epochs or restart conditions should be explicit. An *attempt* is one execution of a plan id over a predeclared epoch/window; retries, aborts, and detector changes are attempts and are logged. A *manifest* is a signed, hash-addressed statement committing code hash, plan id, parameters, and a randomness binding, and pointing to an offline artifact bundle (“available on request”).

Why this matters now. Even mature, widely deployed P2P systems continue to see practical deanonymization and routing-layer censorship, underscoring how small, systematic artifacts can collapse anonymity claims if left unmeasured.[27, 29]

Related work: anonymous DHTs and lookup leakage. Structured-overlay anonymity systems such as Salsa [30] and AP3 [31] use DHT lookups to locate relays, and follow-on analyses show that lookup mechanisms can leak and be attacked [32, 33]. Several secure/anonymous DHT designs therefore include *inspection/testing* queries and related defenses [26, 28]. Our contribution is not a new inspection mechanism, but a general evaluation contract: we bound audit-live distinguishability on a published trace schema so that any trace-based metric transfers to deployment up to an explicit, monitored advantage budget.

2 Audit awareness as distribution shift

Audit awareness is a special case of a broader measurement failure mode: if probes are distinguishable from organic traffic, an adversary (or incentivized intermediary) can treat probes preferentially and bias reported performance. This problem is well-studied in Internet path-quality measurement [34, 35] and appears prominently in the censorship evasion arms race, where censors actively detect and react to measurement/evasion attempts [10]. In anonymous/secure DHT work, the same concern shows up as a design requirement that “testing” or “inspection” lookups be indistinguishable from ordinary lookups, otherwise attackers can behave honestly only on probes [26, 28, 32].

Let Z be the evaluator’s observable trace for a lookup session: timing buckets, hop count, message counts, termination time, etc. Let P_{live} be the distribution of Z under deployment traffic

and P_{audit} under the evaluator’s probes. An evaluation-aware system can implement a policy $\pi(z)$ that behaves differently on audit traces, e.g., serve probes honestly but bias or leak under live traffic.

2.1 Advantage and the contract

Define the audit advantage (TV distance)

$$\text{TV}(P_{\text{audit}}, P_{\text{live}}) = \sup_A |P_{\text{audit}}(A) - P_{\text{live}}(A)| = \frac{1}{2} \sup_{\|f\|_\infty \leq 1} |\mathbb{E}_{P_{\text{audit}}}[f(Z)] - \mathbb{E}_{P_{\text{live}}}[f(Z)]|.$$

Equivalently, under equal priors the Bayes-optimal audit detector has accuracy Acc^* and advantage $\text{Adv} := 2\text{Acc}^* - 1$, and one has $\text{TV}(P_{\text{audit}}, P_{\text{live}}) = \text{Adv}$. This is exactly the largest possible audit-live bias for any bounded statistic of the trace.

Remark (hypothesis-testing privacy). The advantage contract is a hypothesis-testing constraint on the audit vs. live trace distributions. Differential privacy has a closely related “tradeoff function” formulation (f -DP), of which Gaussian differential privacy (GDP) is a one-parameter family; these languages provide composable accounting for trace randomization when desired.[18]

Theorem 1 (Advantage contracts bound evaluation bias). *For any measurable g with range in $[0, 1]$,*

$$|\mathbb{E}_{\text{audit}}[g(Z)] - \mathbb{E}_{\text{live}}[g(Z)]| \leq \text{TV}(P_{\text{audit}}, P_{\text{live}}).$$

Therefore, if a deployment discipline enforces $\text{TV}(P_{\text{audit}}, P_{\text{live}}) \leq \varepsilon$, then any bounded evaluation metric computed from traces (risk, leakage proxy, success rate) is biased by at most ε .

Remark (range scaling). If g has range in $[a, b]$ then applying Theorem 1 to $(g - a)/(b - a)$ gives $|\mathbb{E}_{\text{audit}}[g] - \mathbb{E}_{\text{live}}[g]| \leq (b - a) \text{TV}(P_{\text{audit}}, P_{\text{live}})$.

Proposition 1 (Schema-preserving reuse of an advantage contract). *Let Z be the published trace schema on which an advantage contract establishes $\text{TV}(P_{\text{audit}}^Z, P_{\text{live}}^Z) \leq \varepsilon$. If a later certificate uses a schema $Z' = \phi(Z)$ that is an explicit measurable coarsening of Z , then the same contract transfers conservatively:*

$$\text{TV}(P_{\text{audit}}^{Z'}, P_{\text{live}}^{Z'}) \leq \varepsilon.$$

If the later certificate changes the schema in any other way, the old contract should not be presented as the same claim; the deployment should publish a new exposure-normal-form id and a fresh advantage contract.

Proof. When $Z' = \phi(Z)$ is a coarsening, the law of Z' is the pushforward of the law of Z . Total variation cannot increase under measurable maps, so the bound follows by data processing. For non-coarsening schema changes there is, in general, no relation between the old and new distinguishability problems, so reusing the old contract would compare different observables rather than one continuing claim. $\square$

Proposition 2 (Composing stealth with an audited anonymity certificate). *Let P_0 be an “ideal” trace distribution for the anonymity objective (as in the earlier series papers), and let P_{audit} and P_{live} be the audit and live trace distributions. If an audited certificate establishes $\text{TV}(P_{\text{audit}}, P_0) \leq \varepsilon_{\text{cert}}$ and the advantage contract enforces $\text{TV}(P_{\text{audit}}, P_{\text{live}}) \leq \varepsilon_{\text{adv}}$, then the corresponding live (deployment) trace leakage satisfies*

$$\text{TV}(P_{\text{live}}, P_0) \leq \varepsilon_{\text{cert}} + \varepsilon_{\text{adv}}.$$

Field	Why it matters
Published trace schema id/version	Defines what detectability and bias are measured on (and what detectors are “covered”).
Exposure normal-form id	Stable identifier / schema hash for the exact transcript projection covered by the advantage contract; this is what later certificates must match (or explicitly coarsen) to claim continuity.
Epoch / restart rule	Prevents pooled estimates from hiding high-detectability windows; defines the validity interval of ε .
Audit selection rule	Makes probes unpredictable before execution and auditable after (e.g., commit + beacon/VRF).
Detectability statistic	Specifies the detector/score used for monitoring (e.g., threshold-Youden on a fixed score (a conservative TV lower bound)); enables recomputation.
Detector recipe id	Hash commits features/model class/training pipeline to prevent post-hoc detector tuning; changes imply a restart.
Lineage relation (optional)	Predecessor contract id plus relation tag (same-enf , coarsening , new-interface) when this contract is compared to an earlier one.
Budget ε	The enforced advantage threshold (TV budget) for transfer of evaluation claims.
Publication pointers	Hash/pointer to trace archive plus notarized log entry and inclusion evidence as in the companion release-notary note [17].

Table 1: A minimal, artifact-light “advantage contract certificate” template.

Thus, an audited trace-level anonymity certificate transfers to deployment up to an additive stealth budget.

2.2 Reader verification checklist (minimal, artifact-light)

A published advantage contract can remain small while still being referee-checkable. At minimum, publish (or link to) a compact bundle containing: the trace schema id/version, the exposure normal-form id, the declared epoching/restart rule, the detector recipe id, the claimed advantage budget ε (or a confidence bound for it), and the notarized selection/manifests for the audited epochs as in [17].

A referee can then verify, mechanically:

1. **Schema coverage.** The published trace schema plausibly covers the detectors the system could use to distinguish audits from live traffic (or else ε is explicitly a lower bound).
2. **Lineage discipline.** If the contract is advertised as an update of an earlier one, verify that the exposure normal-form id is unchanged or explicitly declared as a coarsening, as required by Proposition 1; otherwise treat it as a new contract.
3. **No post-hoc detector tuning.** The detector recipe id was fixed before evaluation; detectability is reported on held-out epochs/splits under the stated recipe (or changes are logged as restarts).
4. **Unpredictable probe selection.** Audit selection is commit–reveal/VRF/beacon-derived under the logged commitments, so probes could not be pre-targeted or post-hoc cherry-picked.

Figure omitted in source-only release. Requested file:
figures/advantage_contract_bias_bound.png.

Figure 1: Illustration: bounding trace-level audit advantage controls worst-case bias in bounded metrics.

5. **Restart accounting.** Epoch boundaries and restarts are explicit and consistent with the certificate bundle; high-detectability windows are not averaged away.

This is enough for readers to treat the stated ε as an auditable transfer budget (Theorem 1) without publishing full artifacts inline.

Interpretation for anonymous DHTs. If your anonymity certificate is computed from trace observables, then *stealth* is the missing assumption: audits must be distributionally close to live traffic, or evaluation-aware nodes can make certificates meaningless. Moreover, the transfer statement is only as stable as the declared observable: if the covered trace schema changes, the deployment should publish a fresh exposure id and a new advantage contract rather than silently treating the new monitor as “the same” one.

Figure 1 illustrates the bias bound graphically in a toy model.

3 Monitoring detectability with anytime-valid evidence

We use anytime-valid evidence (e-values / confidence sequences) to reason about *detectability* of audits under adaptive behavior. This paper treats those tools as *black-box primitives*: they ensure that a monitor who decides *when* to look (or when to alert) does not accidentally inflate false positives.

Pointer. The full “anytime-valid spending” story for repeated attempts, optional stopping, and certificate publication is developed in Evaluation Series Paper 1 (Notarized Anonymity Certificates Under Retries). Here we only need one consequence: if the monitor publishes a nonnegative e-value stream, then adversarially chosen stopping times do not invalidate the declared detection threshold.

What is new here. We focus on how an *audit-aware* adversary can shift distributions to evade detection, and on how to pick stealth probes (slice selection) so that detection remains possible without revealing the probe family.

4 Leakage discovery at scale: slice libraries without log-factor blowup

Anonymous DHT systems have many potential leakage slices (key-prefix regions, latency regimes, AS paths, churn states, etc.), which creates a many-testing problem. A practical default is to compute e-values per pre-registered slice and apply the e-BH procedure [4] to control FDR under arbitrary dependence without the strong log-factor penalties of BY-style corrections [5]. For sequential “dashboard” use, there is a subtle filtration issue: repeatedly re-running e-BH under global stopping can fail without additional conditions [19]; see also [16]. For space, the short version omits the full dashboard protocol and toy power comparisons;

Figure omitted in source-only release. Requested file: `figures/slice_hiding_audit_leakage.png`.

Figure 2: Toy slice-hiding effect: predictable audits can be routed around, increasing audit-live bias.

5 Stealth mechanisms: slice-hiding and verifiable probe selection

Even if the auditor samples probes from a realistic traffic model, an evaluation-aware overlay can route around *predictable* probes (slice-hiding). A practical mitigation is to make the set of audited keys/sessions unpredictable until after execution, but verifiable after the fact.

5.1 Commit–reveal with public randomness

A minimal pattern is:

1. commit to an audit pool (e.g., a seed commitment or a hash of candidate probe descriptors),
2. derive the realized audit sample using a public randomness beacon [8, 20, 9, 21] or a VRF [7] (optionally hash-mixing multiple beacons for robustness),
3. reveal and log the selection so anyone can verify it was not post-hoc chosen.

The aim is not secrecy but *unpredictability before execution* and *auditability after execution*. To keep this verifiable under retries and selective disclosure, treat each initiated selection attempt (including aborts) as a notarized, log-appended manifest entry, and budget restarts with the δ -spending discipline in the companion notarization note [17]. Similar “probe resistance” concerns arise in censorship measurement and in DHT crawling/monitoring infrastructure, where measurement traffic can be fingerprinted and routed around. IPFS in particular has been repeatedly crawled and monitored in measurement studies, reinforcing that stable audit infrastructure is realistic (and therefore potentially detectable) in deployed Kademlia-family overlays [15, 14]. We treat these as close analogues of audit-aware behavior and cite both the censorship setting and the P2P setting [11, 10, 12, 13]. Figure 2 illustrates slice-hiding under predictable audits; the extended version includes a verifiable hidden audit selection (VHAS) toy protocol.

6 Padding filters as “stealth actuators”

To reduce detectability, the auditor (or system) can apply a padding/actuation filter that maps audit traces toward the live distribution. This is precisely the type of mechanism treated by the series padding compiler and transport/dual-certificate work [1, 2]: choose a kernel (delay padding, batching, dummy queries) that minimizes overhead subject to a divergence/TV constraint.

In the extended version we include an example of a simple LP-based latency equalization filter from the research archive.

7 Limitations and open problems

From AUC to TV. AUC is a ranking metric and does not directly upper/lower bound TV without additional assumptions. However, any concrete score $S = h(Z)$ yields a conservative *lower bound* on trace-level TV via the best threshold’s Youden statistic (Section 3.2), which can be estimated on held-out data. Tight, dependence-robust conversions (including confidence sequences for these bounds) remain useful work. If one is willing to replace classical TV with a smoothed/blurred

variant, recent work gives distribution-free upper and lower confidence bounds that may be practical for high-dimensional continuous trace features [22].

Schema coverage. The advantage contract only constrains audit–live distinguishability on the *published trace schema* Z . If audit traffic differs on unlogged side channels (e.g., IP ranges, transport fingerprints, lower-layer pacing) or if nodes condition on features omitted from Z , then the contract does not bound evaluation bias for metrics that depend on those omitted features. For high-assurance claims, the publication should justify why the released schema captures the relevant audit detectors, or treat the advantage budget as a lower bound.

Dependence and churn. Anonymous DHT traces are dependent and nonstationary (peer reuse, caching, churn). Anytime-valid tools remain useful, but audit/live modeling and slice definitions must be revisited under major regime change. A conservative policy is to treat configuration or membership shifts as restarting a new monitoring epoch.

DHT-native stealth mechanisms. The case study is intentionally lightweight; a richer simulator (e.g., Kademlia-like routing with churn and latency models [6]) would strengthen the engineering story and can be integrated without changing the theoretical contract.

8 Takeaways

- **Evaluation pipelines are gameable:** if the system can detect audits/probes, it can selectively behave and inflate “privacy budget” reports without improving real privacy.
- **Advantage contracts separate model from measurement:** define the evaluator’s advantage in distinguishing honest from gaming behavior, then design audits to drive that advantage to negligible.
- **Stealth and public randomness are first-class:** commit–reveal, randomized reveal order, and notarization patterns reduce cherry-picking and last-revealer attacks in evaluation itself.
- **Non-redundancy:** treat retry discipline and transcript projections as imports (Operational A; Synthesis 3) and focus this paper on audit-aware evaluation design.

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